Urinary cadmium in the 1999-2008 U.S. National Health and Nutrition Examination Survey (NHANES).

Chronic low-level cadmium (Cd) exposure is linked to kidney and cardiovascular disease, fractures, and cancer. Diet and smoking are primary sources of exposure in the general population. We analyzed urinary Cd in NHANES 1999-2008 to determine whether levels declined significantly over the decade for U.S. children, teens, and adults (nonsmokers and smokers) and, if so, factors influencing the decline(s). For each subpopulation, we modeled log urinary Cd using variable-threshold censored multiple regression. Models included individual-level covariates (age, gender, BMI, income, race/ethnicity/country of origin, education, survey period), smoking, housing (home age, water source, filter use), and diet (supplement use; 24-h calorie, fat, protein, micronutrient, and Cd-containing food intakes), creatinine, and survey year variables. Geometric mean urinary Cd (ng/mL) declined 20-25% in these subpopulations, and the regressions showed statistically significant declines in later years for teens and adults. While certain covariates were significantly associated with Cd by subpopulation (creatinine; age; BMI; race/ethnicity/origin; education; smokers in the home; serum cotinine; 24-h fat, Mg, Fe intakes; use of dietary supplements), they did not help explain the declines. Instead, unidentified time-related factors appeared responsible. Despite the declines, millions of Americans remain potentially at risk of adverse outcomes associated with low-level Cd exposure.